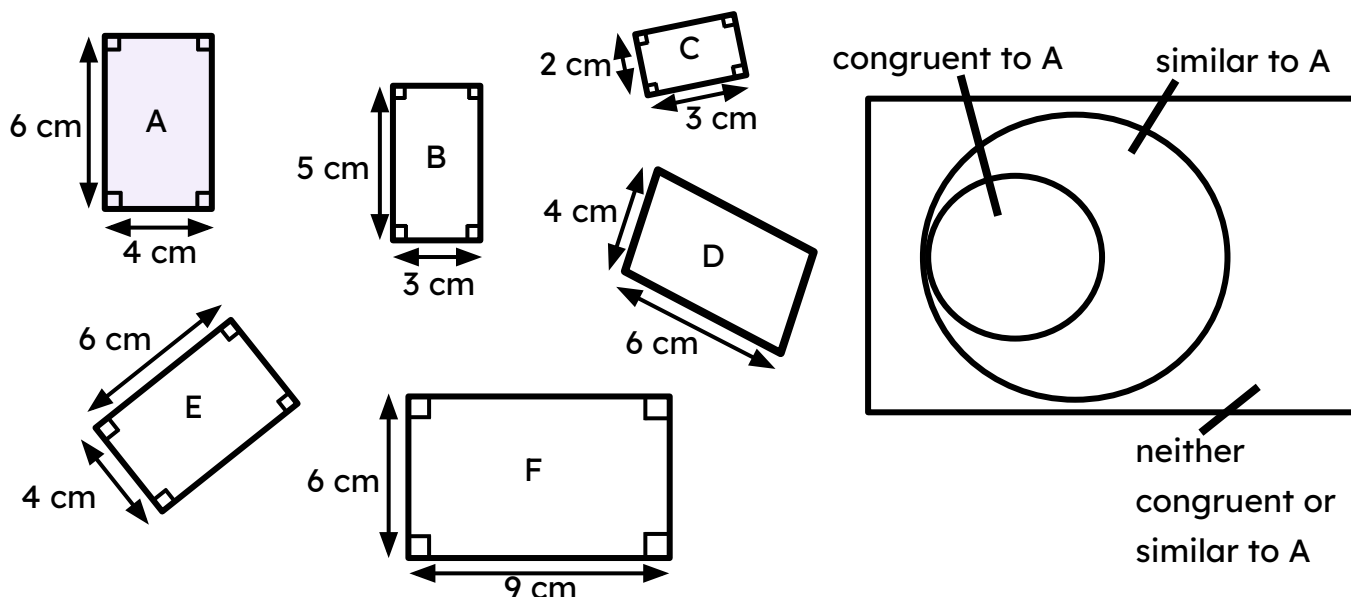




## Congruent or similar or neither

### Task A: Distinguishing between similarity and congruence

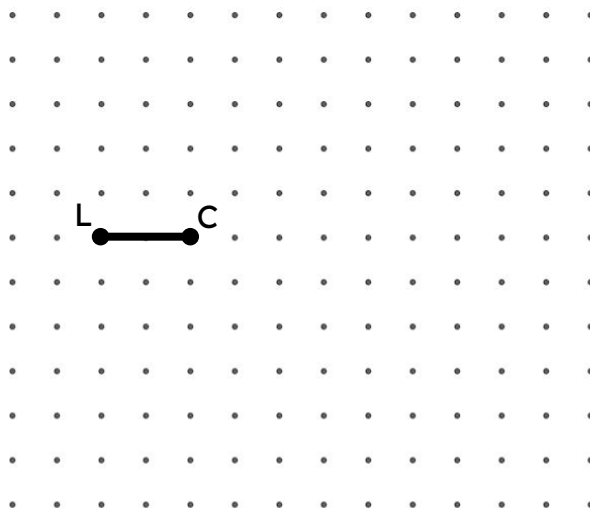
1) Compare each shape with Shape A and write the letters B to F in the Venn diagram.



2) The information describes five connected rectangles, with vertices at points A to L.

- LC is a horizontal line with length 2 units.
- Point F is below Point C and  $CF = 2LC$ .
- LCFI is a rectangle.
- KLIJ is congruent to LCFI.
- ABCL is congruent to LCFI.
- CDEF is similar to LCFI but not congruent.
- FGHI is similar to LCFI but not congruent.

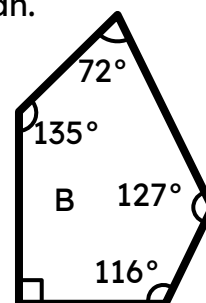
Use the information to plot points A to L on the grid and draw the five rectangles.



## Task B: Checking for similarity and congruence

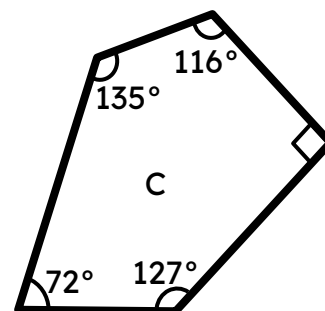
1) a ) Fill in the table for shapes B and C, matching up the angles if you can.

|                       |      |      |     |      |     |
|-----------------------|------|------|-----|------|-----|
| <b>shape A angles</b> | 116° | 127° | 72° | 135° | 90° |
| <b>shape B angles</b> |      |      |     |      |     |
| <b>shape C angles</b> |      |      |     |      |     |



b) Which shape is neither similar nor congruent to Shape A?

c) Is it possible to determine congruence for the other shape? Explain your answer.



2) a ) Fill in the table for shape B.

|                            |     |      |      |     |
|----------------------------|-----|------|------|-----|
| <b>angles</b>              | 42° | 109° | 119° | 90° |
| <b>shape A lengths (m)</b> | 6   | 4    | 2    | 8   |
| <b>shape B lengths (m)</b> |     |      |      |     |

b) Are shapes A and B congruent, similar or neither?  
Justify your answer.

